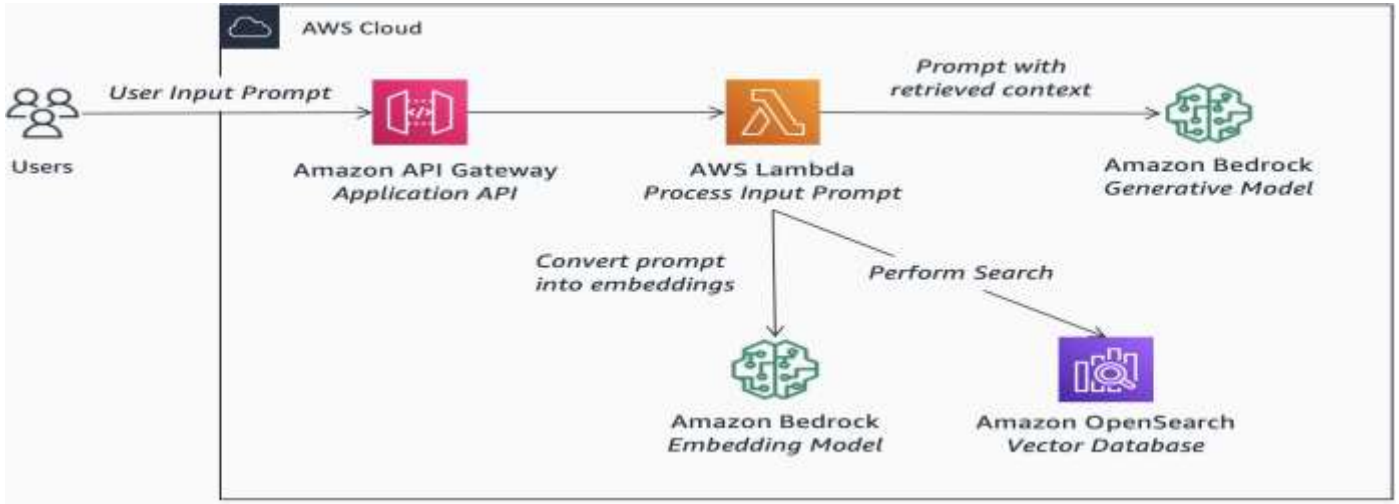
Amazon Bedrock Optimization, System Performance & Cross-Region Inference

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AMAZON BEDROCK OPTIMIZATION – INTRODUCTION

Security Services

- Amazon Bedrock
- Foundation Models
- Inference optimization
- System performance
- Scalable AI applications



Description

This presentation explores optimization techniques for Generative AI applications built with Amazon Bedrock. It covers use-case-specific optimization, Foundation Model performance, request handling, and cross-Region inference.

OPTIMIZING FOR SPECIFIC USE CASES

Main Features

- Application requirements
- Task complexity
- Model capabilities
- Performance expectations
- Cost considerations



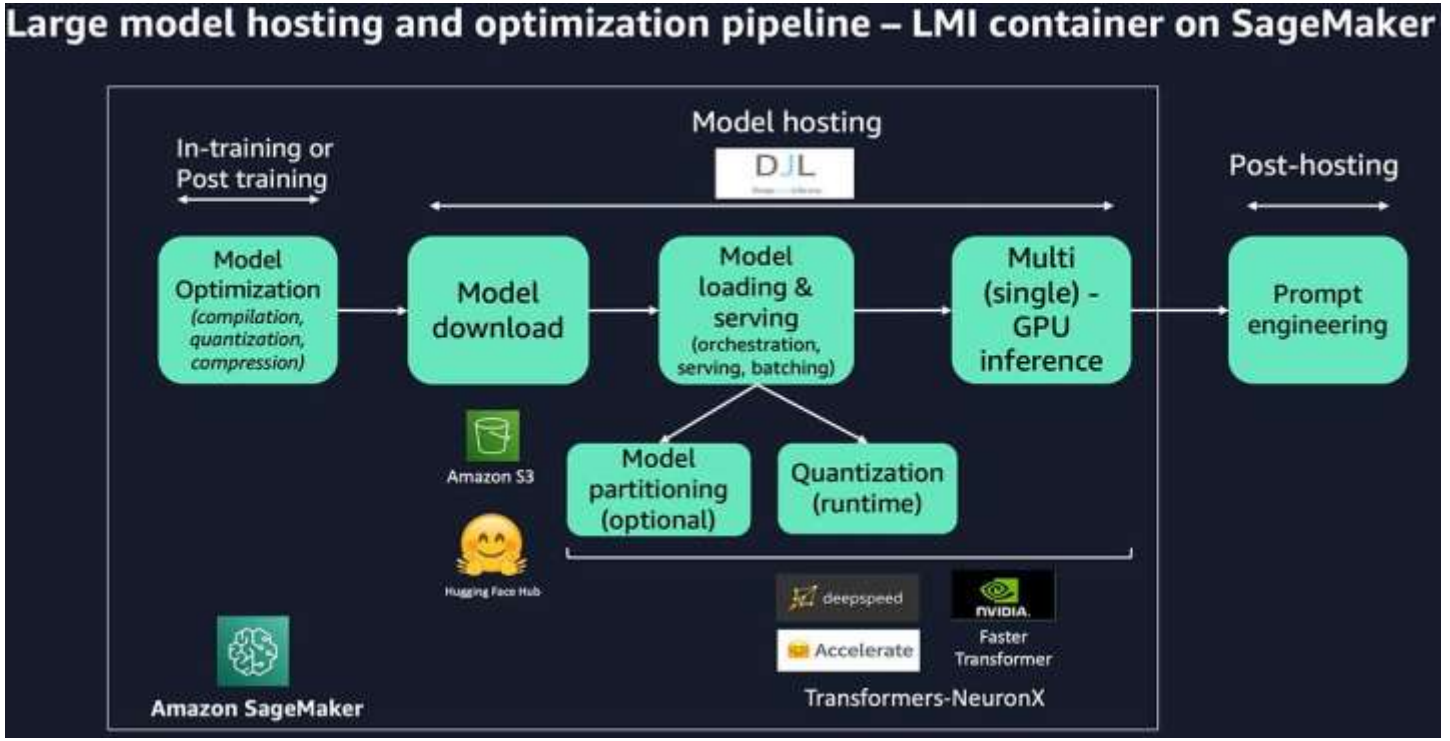
Description

Optimization decisions should reflect the requirements of the application. Developers need to consider task complexity, model capabilities, latency expectations, cost, and response quality when selecting and configuring a Generative AI solution.

OPTIMIZING FOUNDATION MODEL SYSTEM PERFORMANCE

Main Features

- Inference latency
- Throughput
- Request processing
- Model configuration
- Performance monitoring



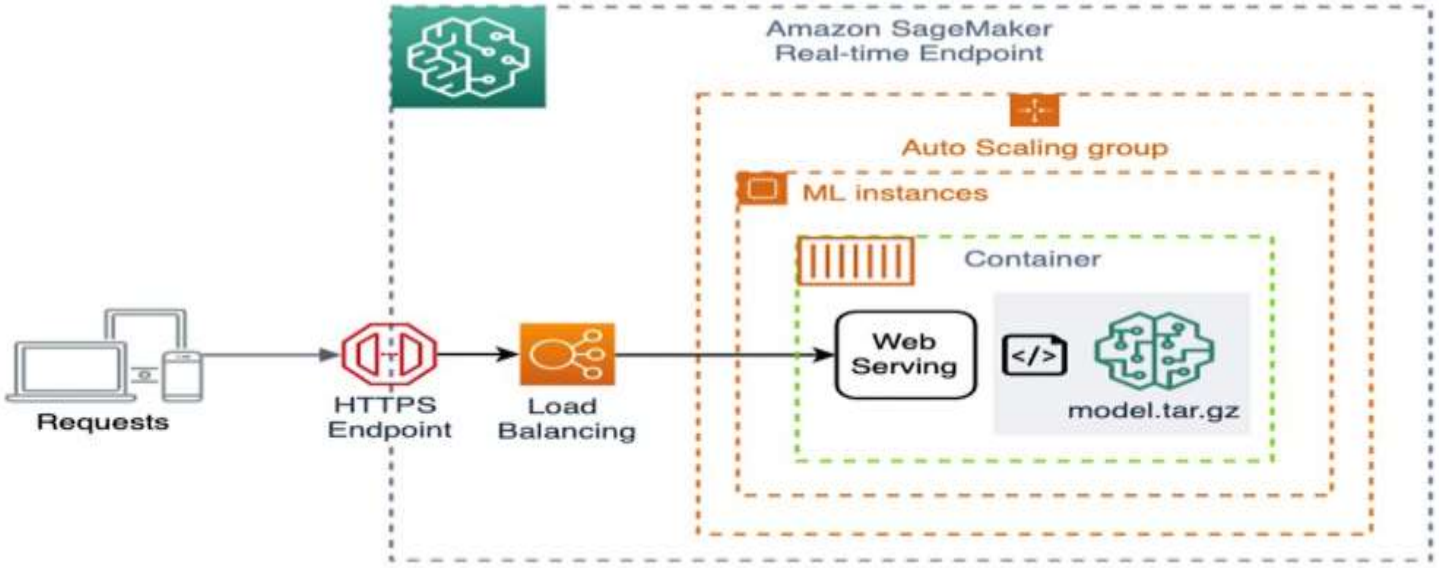
Description

Foundation Model system performance depends on factors such as request volume, model capabilities, inference configuration, and application design. Monitoring these factors helps developers identify performance bottlenecks and opportunities for improvement.

SCALING AI INFERENCE WORKLOADS

Main Features

- Concurrent requests
- Workload distribution
- Request management
- Service quotas
- Scalability



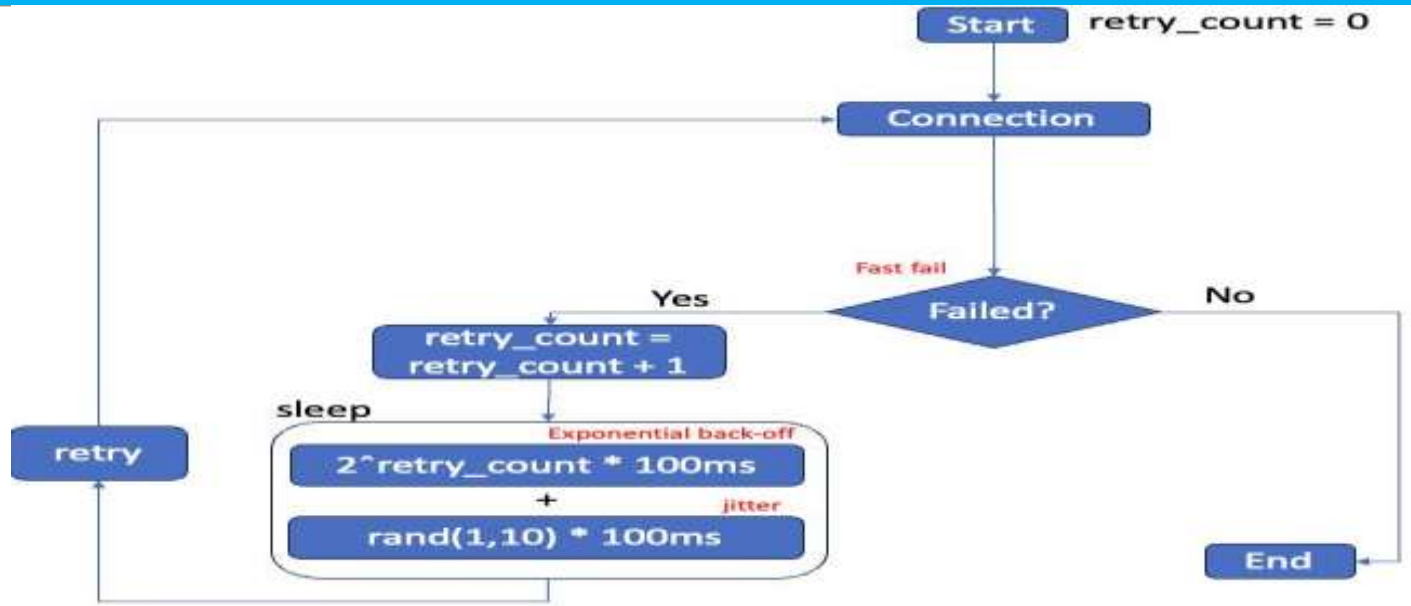
Description

Scaling AI inference workloads involves managing requests as application demand changes. Developers should understand service quotas, supported concurrency, and request-handling behavior to design applications that meet their performance requirements.

EXPONENTIAL BACKOFF AND CONNECTION POOLING

Main Features

- Retry delays
- Exponential backoff
- Transient errors
- Connection reuse
- Resilient API requests



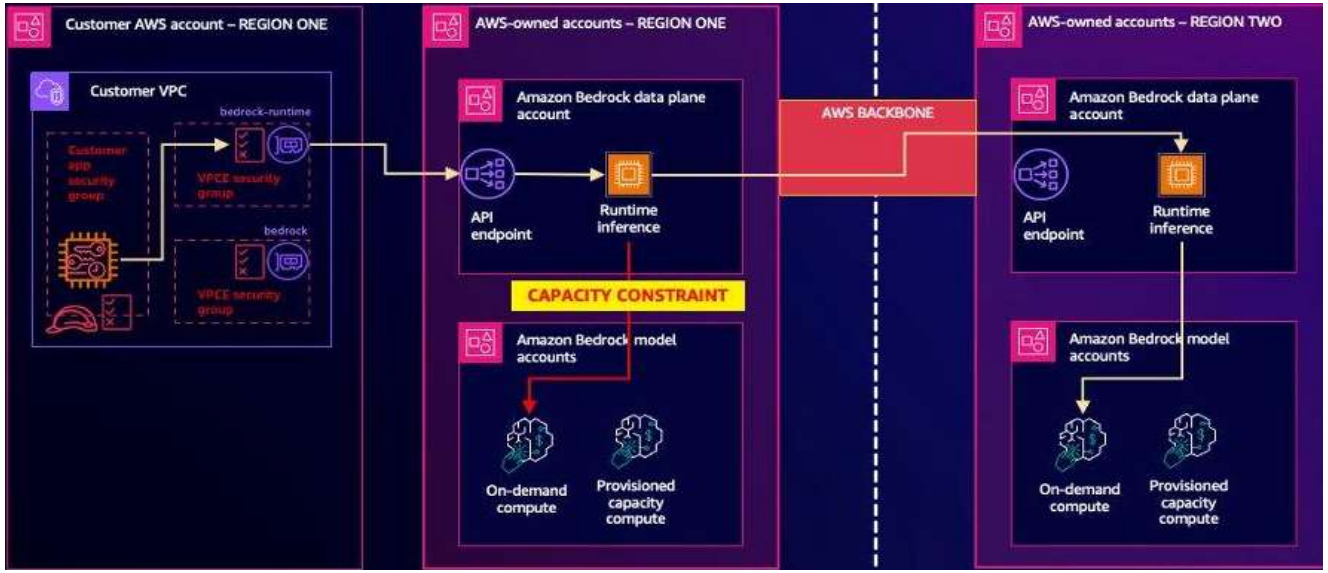
Description

Exponential backoff helps manage transient errors by increasing the delay between retry attempts. Connection pooling reduces the overhead of establishing new connections by reusing existing ones where supported. These techniques can improve the reliability of applications that interact with AI services.

INTRODUCTION TO CROSS-REGION INFERENCE

Main Features

- AWS Regions
- Model inference
- Inference profiles
- Regional capacity
- Request routing



Description

Cross-Region inference is an Amazon Bedrock capability that can route supported inference requests across eligible AWS Regions using the appropriate inference profile. It can help applications access available model capacity according to supported routing configurations.

AMAZON BEDROCK CROSS-REGION INFERENCE

Main Features

- Cross-Region routing
- Amazon Bedrock
- Inference profiles
- Supported Foundation Models
- Regional request processing



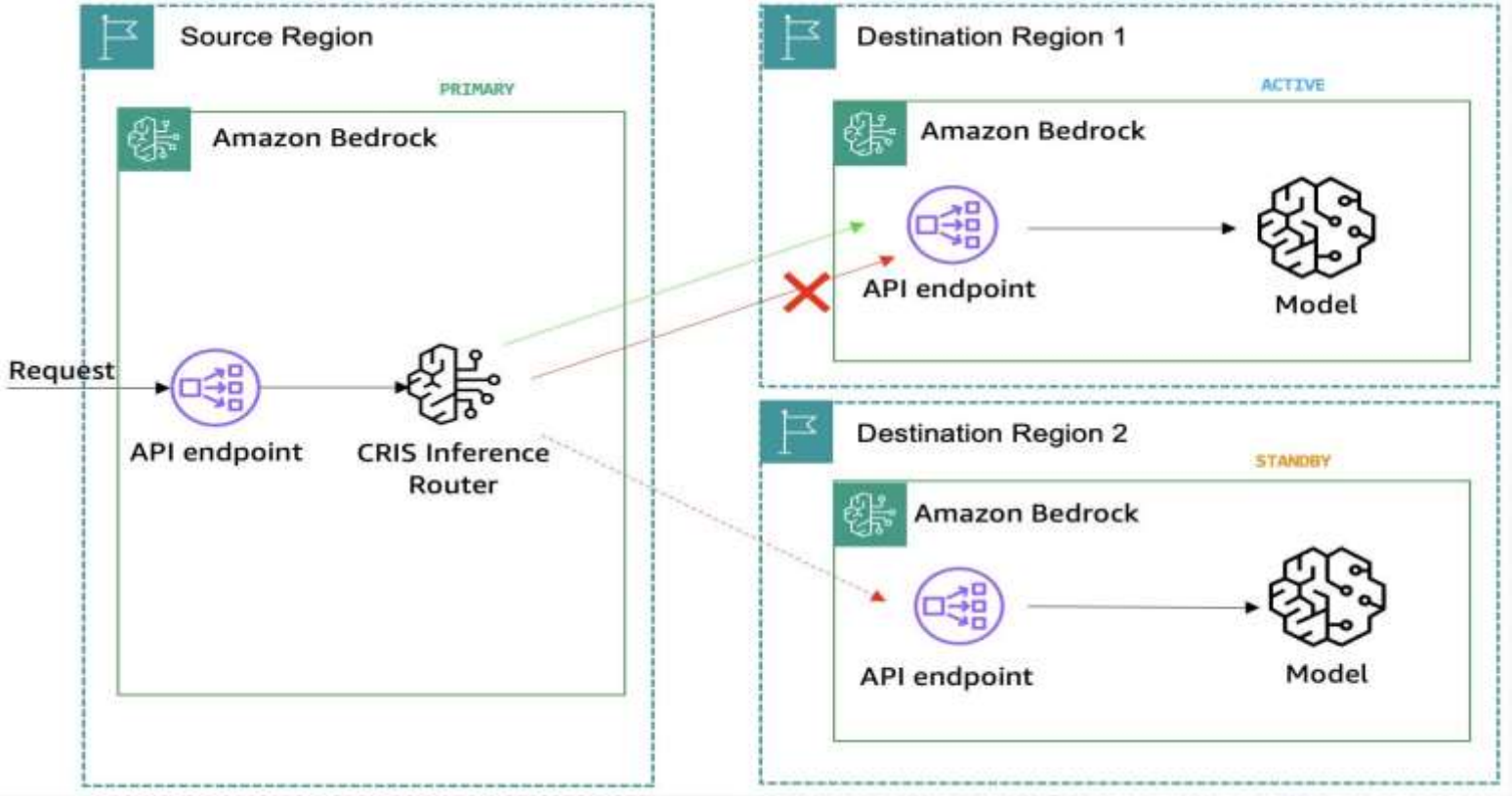
Description

Amazon Bedrock cross-Region inference allows supported requests to be routed across eligible Regions through an inference profile. Developers need to understand the applicable model, profile, and regional support when configuring this capability.

BENEFITS OF CROSS-REGION INFERENCE

Main Features

- Inference capacity
- Workload distribution
- Performance considerations
- Application scalability
- Regional availability



Description

Cross-Region inference can help manage inference demand by allowing supported requests to use capacity across eligible Regions. The actual benefits depend on the selected model, inference profile, supported Regions, and application requirements.